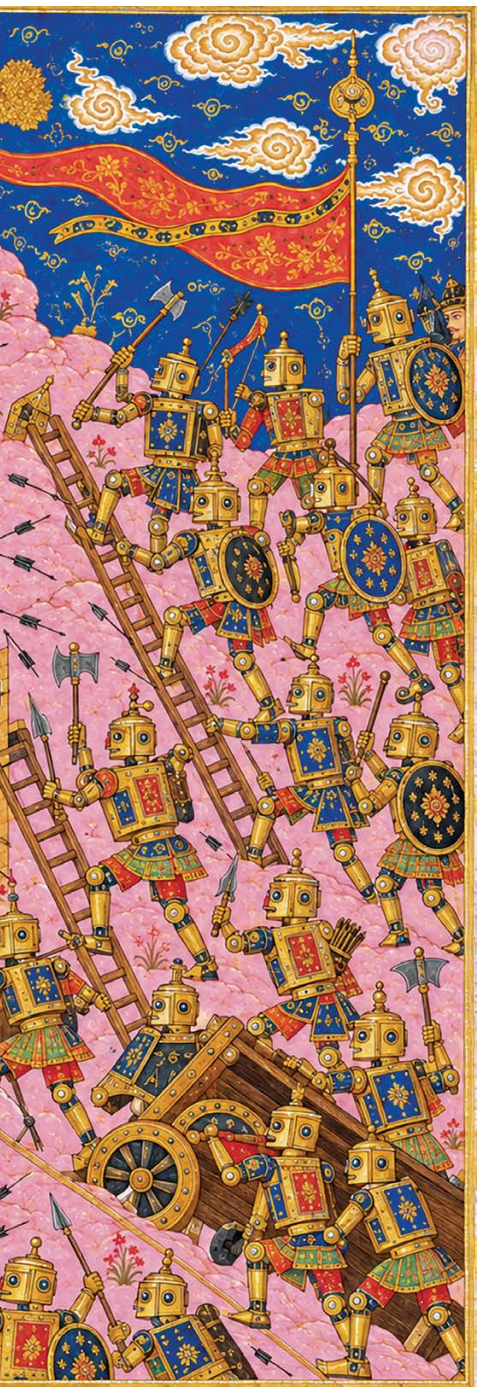


decades to build. With over $100 billion (Rs 9.6 lakh crore) of investment now committed by hyperscalers, co-location providers and domestic companies, capacity should reach 6-8 GW by 2030.

But building capacity is only one part of the equation. As AI moves from experimentation into production, enterprises will increasingly need infrastructure that they can trust to run sensitive and mission-critical workloads. Security, trust, transparency, compliance and control will therefore become equally critical requirements for enterprise AI infrastructure.

This is also where AI sovereignty and AI security begin to converge. Sovereignty in the AI era cannot mean only that compute or data reside within national borders. It also means meaningful control over the data, models, infrastructure—and the ability to independently monitor and secure them.

Second is India's existing technology and cybersecurity base. India is entering this emerging market not from a standing start, but with a highly credible, rapidly expanding base. The country is home to over 400 cybersecurity product companies that generated $4.46 billion (Rs 42,968 crore) in revenue in 2025. Achieving a 34 per cent compound annual growth rate, more than 55 per cent of them have expanded footprint globally. This mature product and talent ecosystem can anchor the next generation of AI security.

The third is India's ability to deploy technology at scale. Global enterprises will need continuous model testing, agent identity management, AI red-teaming, secure deployment, software-supply-chain assurance, vulnerability triage, AI governance and managed detection. This is essentially the next evolution of cybersecurity-as-a-service. There is evidence the market is moving this way already. Indian BFSI organisations are directing incremental cybersecurity spending toward AI, with agents expected to handle more than a quarter of Tier-1 SOC (security operations centre) work within two to three years. The opportunity for India—besides defending domestic systems—is also in building the next generation of cybersecurity services for a world adopting AI at scale.

Cyber services to AI assurance

Every enterprise deploying AI will now need answers to a new set of questions. Can the model be manipulated? Can proprietary data leak through it? What systems can an AI agent access? How do you detect abnormal agent behaviour? How does the system respond when the underlying model changes?

These questions cannot be answered through a one-time audit. AI systems evolve with models, data, prompts, tools and user behaviour. Assurance, therefore, must become continuous. Second, India needs a deeper talent pipeline spanning AI and cybersecurity. The skills will increasingly converge: cybersecurity professionals will need greater understanding of models and machine learning, while AI engineers will need to understand identity, secure architectures, adversarial behaviour and governance. Third, we need shared testing environments and deep collaboration between industry, government, academia and startups. AI models will also have to be tested against real-world operating conditions before deployment.

India has an opportunity to build that trust layer, turn it into a global industry. But this will depend equally on ecosystems that can test that intelligence, secure it, govern it and make it dependable enough for real-world use. ■

TAKEAWAYS

- AI is changing the security perimeter: threats now include model manipulation, data leakage, risky AI-agent behaviour...
- India's data infrastructure, cybersecurity firms, global tech services capability position it well to lead in trusted AI
- AI assurance must be continuous, for systems are evolving with changing models, data, prompts, user behaviour

The author is President, Nasscom